# 5.6.6 Gravitational Force

**After completing this chapter you should be able to apply models, theories and laws to situations involving energy, force and motion.**

**For Questions 1 to 9 choose the best answer (A), (B), (C) or (D).**

**1.** Which statement about the mass and weight of an object is correct?

(A) Both change if the position of an object in a gravitational field changes.
(B) If an object is in a zero gravity environment both its mass and weight are zero.
(C) The mass of an object can be zero, but its weight cannot equal zero.
(D) An object can have zero weight, but its mass can never be zero.

**2.** Which statement correctly defines mass?

(A) Mass is the weight of the matter in an object.
(B) Mass is a measure of the amount of matter in an object.
(C) Mass is a measure of the gravitational force acting on an object.
(D) Mass is a measure of the effect of gravitational acceleration on an object.

**3.** Which choice correctly identifies the units used to measure mass and weight?

| | Mass | Weight |
|---|---|---|
| (A) | Newtons | Kilogram |
| (B) | Newtons | Newtons |
| (C) | Kilogram | Newtons |
| (D) | Kilogram | Kilogram |

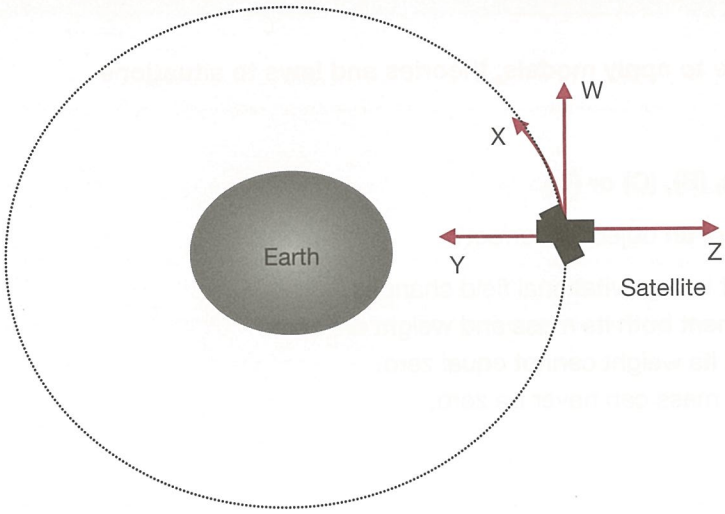
**4.** Which statement about mass and weight is correct?

(A) Your mass would be the same on Earth and Jupiter.
(B) Your weight would be the same on Earth and Jupiter.
(C) Your mass would be less on Earth than on Jupiter.
(D) Your weight would be more on Earth than on Jupiter.

**5.** The force due to gravity on the Moon is about one sixth that of the Earth. On Earth a 10 kg mass has a weight of about 98 newtons. Which choice correctly estimates the mass and weight of the block on the Moon?

| | Mass of block on Moon | Weight of block on Moon |
|---|---|---|
| (A) | 10 kg | 16.3 N |
| (B) | 16.3 kg | 10 N |
| (C) | 60 kg | 98N N |
| (D) | 10 kg | 588 N |

**Use the following information to answer the next THREE questions.**




**6.** Which arrow correctly shows the direction of the force of gravity acting on the satellite?

(A) W
(B) X
(C) Y
(D) Z

**7.** Which of the following does *not* affect the size of the force of gravity acting on the satellite?

(A) The mass of the satellite.
(B) The mass of the Earth.
(C) The altitude of the orbit of the satellite.
(D) The orbital speed of the satellite.

**8.** An identical satellite, M orbits Mars. Both satellites E and M are the same distance from the centre of the planet they orbit. Earth has about ten times the mass of Mars.

Which statement correctly compares the gravitational force on the two satellites?

(A) The force on satellite E is equal to the force on satellite M.
(B) The force on satellite E is greater than the force on satellite M.
(C) The force on satellite E is less than the force on satellite M.
(D) There is no force on either satellite as they are in free space.

**9.** Which statement is correct?

(A) A person on top of Mt Kosciuszko would have less mass than a person at sea level.
(B) A person on top of Mt Kosciuszko would have more mass than a person at sea level.
(C) A person on top of Mt Kosciuszko would weigh less than a person at sea level.
(D) A person on top of Mt Kosciuszko would weigh more than a person at sea level.

**10.** A group of students wants to do an experiment to see if the rate at which objects falls depends on their mass.

(a) Write a hypothesis for this experiment.

objects with larger masses will fall faster than objects with smaller masses.

(b) Suggest how this experiment could be done (you can write a method if you like).

Drop objects of different masses from the same height and record the time it takes each to fall. To increase reliability, repeat each measurement at least 3 times. Repeat at 3 different heights.

(c) List the equipment they would need.

Stopwatch
Balance
metre ruler

(d) What factors would need to be controlled in this experiment?

Shape of object (plasticine ball)
composition of object

(e) What would have to be measured in the experiment?

Height dropped from, mass of object, time to fall to the ground.

(f) Predict the results the students might get.

objects dropped from equal heights take the same time to fall to the ground.

(g) On the basis of your prediction, write a conclusion for the experiment.

The rate of fall of objects of similar shape and composition is the same regardless of their mass.

**11.** Compare the mass and weight of an object.

Mass is the amount of matter contained in kg.
Weight is the force of gravity acting on that mass in N.
Mass is the same regardless of position.
Weight will change depending on the strength of gravity at that position.

**12.** An 60 kg astronaut on Earth has a weight of about 588 N. On Mars her mass will still be 60 kg but her weight will only be about 220 N.

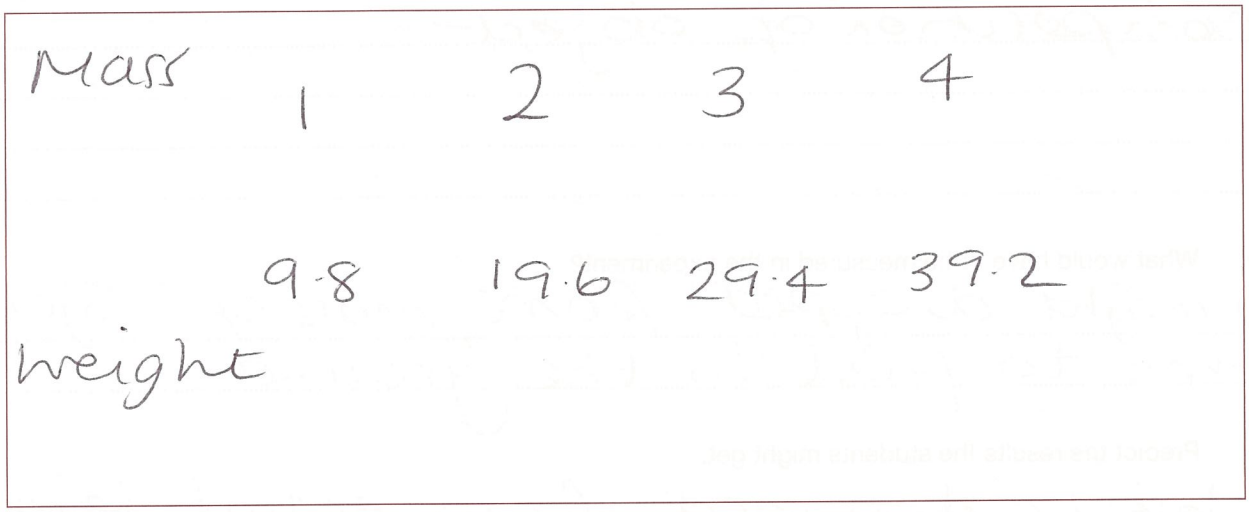
(a) Account for the mass being the same.

Mass is the same regardless of position.

(b) Account for her weight being different.

Weight is less on mars because the strength of gravity on mas is less.

**13.** Explain how the same balance could be used simultaneously to measure mass and weight. Use a diagram to illustrate your answer.



**14.** The weight of an object is given by the following equation:

$W = mg$ where $W$ = weight in newtons

$M$ = mass in kilograms

$g$ = acceleration due to gravity

= 9.8 m $s^{-2}$ at sea level at the equator

Calculate the missing quantities in the table.

| Mass of object on Earth (kg) | Weight of object at sea level on Earth (N) |
|---|---|
| 0.25 | 2.45 |
| 0.60 | 5.88 |
| 2.4 | 23.52 |
| 7.5 | 73.5 |
| 0.51 | 5.0 |
| 3.0 | 29.4 |
| 200 | 1960 |
| 750 | 7350 |

**15.** Using the same equation as in Question 14, but different values for the acceleration due to gravity depending on where the masses are, calculate the missing values in the table.

| Mass of object (kg) | Weight of object (N) | Where object is | Gravity here compared to Earth (Earth = 9.8 =1) |
|---|---|---|---|
| 2.0 | 7.25 | Mercury | 0.37 |
| 5.0 | 43.12 | Venus | 0.88 |
| 4 | 14.9 | Mars | 0.38 |
| 25 | 644.35 | Jupiter | 2.63 |
| 40 | 1030.96 | Jupiter | 2.63 |
| 0.5 | 5.63 | Saturn | 1.15 |
| 15.0 | 171.99 | Uranus | 1.17 |
| 6.0 | 70.54 | Neptune | 1.20 |
| 24.5 | 732.55 | Sun | 29.9 |